



Recent advances in Deep learning based automatic sign language processing



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September 2026

**Image processing
& Embedded Lab**



보이스 인공지능 연구소 Voice AI Research Center

Research interests

AI-driven HCI

Computer Vision

Pose estimation

Action recognition

Sign language recognition

Current position

Assistant Professor, Researcher

Image Processing & Embedded
Lab, Department of Computer
Engineering, Gachon
University

Since 2026.03 -

Background

Senior researcher, Inha
University, 2025.10-2026.02

Postdoc., School of Computing,
KAIST 2025.03 -2025.10

Phd researcher at Xvoice AI Lab,
Inha University, 2019-2025.02

Professor at TUIT, Uzbekistan
2013-2018

Lecturer at DPI, Uzbekistan 2010-
2013

Currently working projects

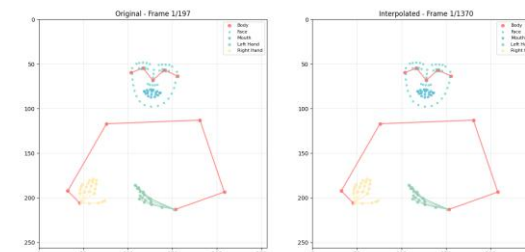
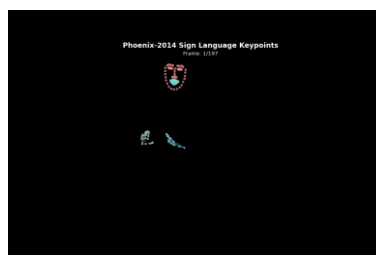
Accessible technologies

Virtual keyboard

Object detectors

Vision models

Web Programming (K0126002)



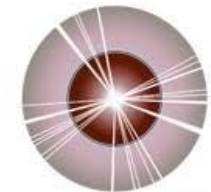
Awards

- **Outstanding Researcher Award (2025)**
 - Inha University recognition for groundbreaking research contributions.
- **Best Researcher Award (2025)**
 - Graduate School's recognition for most presented international papers.
- **Best Paper Award (2024)**
 - Awarded by BigDas Inter. Conf. for exceptional research paper.
- **Best Paper Awards (2024)**
 - Awarded by the Korean Computer Congress for great research paper.
- **Best Paper Awards (2020-2023)**
 - Awarded by BK21, BMSE, Inha University for outstanding research papers.
- **Award the Scientific Title of Docent (2018)**
 - Awarded by the Supreme Attestation Commission of Uzbekistan.
- **Best Researcher Professor Award (2016)**
 - Recognized for exceptional teaching contributions in Tashkent, Uzbekistan.



O'ZBEKISTON RESPUBLIKASI
OLIY TA'LIM, FAN VA INNOVATSIYALAR
VAZIRLIGI

충북대학교
블록체인연구센터



한국정보과학회
KOREAN INSTITUTE OF INFORMATION SCIENTISTS AND ENGINEERS

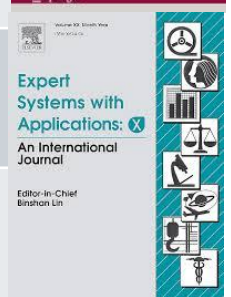


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INHA UNIVERSITY

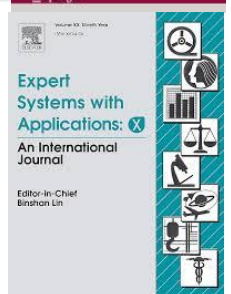
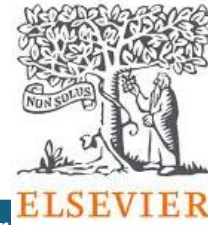
Publications



N	Date	Title	Publisher	IF	Author
1	2026.08	LITE and PRIME: Robust Streamlined MOT Solutions for Real-Time Systems	IEEE Access	4.82	Jumabek Alikhanov, Seunghyun Ko, Mukhiddin Toshpulatov, Hakil Kim,
2	2026.02	Adaptive feature refinement for texture-preserving single image super-resolution	Cluster Computing, Springer US	4.2	Toshpulatov, M., Safarov, F., Lee, G.
3	2025.09	Hyperspectral Anomaly Detection with Enhanced Spectral Graph Transformer Network	IEEE Access	4.82	F Safarov, M Toshpulatov, W Lee et al.
4	2025.02	Deep learning pathways for automatic sign language processing	Pattern recognition	9.84	Mukhiddin Toshpulatov, Wookey Lee, Jaesung Jun, Suan Lee
5	2024.07	DDC3N : Doppler-Driven Convolutional 3D Network for Human Action Recognition	IEEE Access	4.82	M Toshpulatov, W Lee, S Lee, H Yoon, U Kang
6	2023.02	Talking human face generation: A survey	Expert Systems with Applications	9.29	M Toshpulatov, W Lee, S Lee



Publications



N	Date	Title	Publisher	IF	Author
5	2022.01	Human pose, hand and mesh estimation using deep learning: a survey	The Journal of Supercomputing	<u>3.96</u>	M Toshpulatov, W Lee, S Lee, A H. Roudsari
6	2021.03	Generative adversarial networks and their application to 3D face generation: A survey	Image and Vision Computing, Elsevier	<u>4.3</u>	Mukhiddin Toshpulatov, Wookey Lee, Suan Lee



Personal info

- Hobbies:
 - Biking
 - Jogging
 - Exercising(Gym)
 - Inline skating
 - Swimming
- Watch movies
- Reading news, books, and research papers
- In Korea
 - Since 2018.08
- Married:
 - Wife
 - Son
 - Daughter
 - Son



Q & A

muhiddin@gachon.ac.kr

- Please introduce yourself briefly!
 - Your first(or short) name;
 - Country of origin;
 - Department, course, grade, year;
 - Research interests or other fields, hobbies, ...
 - The reason for choosing the Web Programming course
 - Expectation from the course
 - Anything you want

Nice to meet you! ' /

Let me introduce myself 😊



Muhiddin

First (or short) name:
Muhiddin

Country of origin:
Uzbekistan 🇺🇿

Department:
Computer Science and Engineering

Course:
Web Programming

Grade, Year:
PhD Student, 3rd Year

Research interests

- Computer Vision
- Deep Learning
- Human Action Recognition
- Spatio-Temporal Data Analysis



Other fields & hobbies

- Interested in AI, Data Science, and Full-Stack Development
- Love playing and watching football ⚽
- Traveling ✈️ • Reading books 📖
- Exploring new technologies 🖥️



The reason for choosing the Web Programming course

- I want to strengthen my web development skills to build impactful and user-friendly applications.
- It is essential for turning ideas into real-world solutions and expanding my research and career opportunities.



Expectation from the course

- Gain strong foundation in both front-end and back-end web technologies.
- Work on practical projects and real-world applications.
- Learn best practices and modern tools.
- Collaborate, share ideas, and grow together!



Anything I want to add

I am excited to learn from all of you, contribute to the class, and make this journey enjoyable and meaningful! Let's have a great semester together! 🚀

Let's learn together! ❤️

Thank you! I'm looking forward to this amazing learning experience. 😊

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